ModelIt!

Student Activity Pack

One Species Changes Everything

How Keystone Species Hold Ecosystems Together

Grade NE | NGSS MS-LS2-4, HS-LS2-6 | Stability and Change

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 What do you think would happen if ALL the bees disappeared from an area?

2 Are all species equally important to an ecosystem, or are some more important than others?

3 What does the word 'keystone' mean in architecture? How might that apply to ecology?

4 Draw what you think a healthy ecosystem looks like vs. a damaged one.

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Keystone Species Population | Habitat Quality | Food Availability | Number of Other Species

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Thriving Colony

High keystone population

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Species Removed

Zero keystone population

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Recovery

Gradually increasing from zero

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Invasive Species Pressure — Non-native species that compete with natives for resources

Human Development — Construction, roads, and agriculture that reduce habitat

Water Quality — Health of the water system — affected by beaver activity and vegetation

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Can removing just ONE species really destroy an entire ecosystem? Use evidence from your simulation to support your answers.

1. **Why are some species called 'keystone' species? What does the architectural metaphor mean?**

2. **Explain the feedback loop in your model — how does biodiversity feed back into habitat quality?**

3. **Why does ecosystem recovery take time even after the keystone species returns?**

4. **Name a keystone species besides beavers and explain why it is critical to its ecosystem.**

5. **How could this model help conservation agencies decide which species to prioritize for protection?**

8 STEM Challenge: Jenga Ecosystem

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YOUR ENGINEERING MISSION

Play a modified Jenga game where each block represents a species. Discover which removals cause the tower to collapse and which ones the ecosystem can survive.

THE SCENARIO

A conservation agency needs to understand which species are most critical to protect. Your team will simulate species removal using a Jenga tower and identify which species, if removed, cause the ecosystem to crash.

Engineering Design Thinking

1. WHICH BLOCKS WILL YOU LABEL AS 'KEYSTONE' SPECIES?
2. HOW WILL YOU RECORD WHICH SPECIES WERE REMOVED BEFORE THE CRASH?
3. CAN YOU PREDICT WHICH REMOVAL WILL CAUSE THE COLLAPSE BEFORE PULLING?
4. WHAT PATTERN DO YOU NOTICE ABOUT WHICH SPECIES ARE SAFE TO REMOVE VS. DANGEROUS?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Conservation Biologists protect endangered species and restore damaged ecosystems. They identify keystone species and design recovery plans for habitats around the world. They earn $55,000–$95,000/year.